

EXAM 7 – SPRING 2012

9. (2 points)

a. (0.5 point)

Briefly describe two advantages that non-parametric smoothing models for claims run-off pattern estimates have over models based on parametric curves.

b. (0.5 point)

Describe the role of the θ parameter in the incremental claim predictor:

$$\eta_{ij} = s_{\theta_i}(i) + s_{\theta_j}(\log(j)) + s_{\theta_j}(j)$$

c. (0.5 point)

Briefly describe the fit of the model from part b. above when θ is at each extreme.

d. (0.5 point)

Identify the resulting models when θ is at each of its extremes.

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Question 9 Sample Answer

Solution 1

- a) Two advantages =
1. Not as rigid of a fit, so they can fit the data better, using a blending of 2 well known models.
 2. Parametric curves may end up showing trends that aren't in the data when extrapolating, or showing non-intuitive increasing development. Non-parametric models can control this.

Solution 2

- a) There are fewer parameters required. Parametric models have been critiqued for being over parameterized. Smoothing models allow you to extrapolate beyond the end of your data.

Solution 3

- a)
1. Allows for bimodal distributions.
 2. Uses empirical data, which may give a closer approximation of the true dist. than the smoothed model especially @ early maturities. Allows for some volatility.

Solution 4

- a) Advantage – you can reduce the number of parameters in your model which can decrease your estimation (parameter) variance.

Disadvantage – The curve may not be able to match the actual shape of the development pattern and may overstate or understate factors.

Solution 5

- a)
- smoothing models respond more to data than parametric curves (at extreme = CL method)
 - Parametric curves may have too few parameters, improved if early dev periods get own wt.

Solution 6

a)

- 1) For non-parametric smoothing models no further assumptions on loss distribution parameters need to be made.
- 2) Moments beyond the first two moments can be estimated for the loss distribution.

Solution 7

a)

1. Non-parametric models are easy to incorporate judgment into selection of reserves. Parametric models will sometimes pick up false signal in the data.
2. Non-parametric models can handle negative development for salvage and subrogation or beneficial case reserve development. Many parametric models do not support negative development (like over dispersed Poisson and Gamma)

Solution 1

- b) θ is the level of smoothing that takes place, trading off between a rigid fit and the CL method.

Solution 2

- b) The θ parameter works as a smoothing parameter between the analytic result (say an over dispersed Poisson resulting in chain ladder estimates) and the Hoerl curve. The model works to fit and smooth simultaneously. It keeps your output from being too jumpy.

Solution 1

- c) When $\theta = 0$, there is no smoothing and the fit is to the data itself. When $\theta = \infty$, the model is completely smoothed and linear.

Solution 2

- c) When θ is small, more dependency is on the emerging data and when θ is large more dependency is on the parametric model.

Solution 1

d) When $\theta = 0$, this is the chain ladder method. When $\theta = \infty$, this is the Hoerl Curve.

Solution 2

d) When $\theta = 0 \Rightarrow$ chain ladder method
When $\theta \rightarrow \infty \Rightarrow$ Hoerl curve

Examiner Comment

For the a. part, a common answer that did not receive credit was the ability to extrapolate the data as extrapolation is not an unique characteristic/advantage of non-parametric smoothing models.

A common incomplete response for the b. part stated that theta was the smoothing parameter, but failed to identify a tradeoff aspect between goodness of fit and smoothing. Stating that theta “credibility weighted” two curves was also not a valid response.

Credit was given in the c. part if the response explained that the two extremes were unsmoothed and fully smoothed. The two extremes resulted in reliance on the empirical data and a smoothed parametric curve was also a common valid response.

Credit for the d. part was only given for stating the Chain Ladder or Hoerl/Gamma curve. Responses that did not mention the extreme values or note which curve went to each extreme received credit, but responses that matched the extreme with the wrong curve did not.